Praneeth Kodavati

cs24btech11034@iith.ac.in | kssmpranith@gmail.com | +91 8828994589 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.46
Tirumala Junior College	Class XII (Board of Intermediate Education, Andhra Pradesh)	2024	98.10%
Aditya Talent School	Class X (Board of Secondary Education, Andhra Pradesh)	2022	95.16%

ACHIEVEMENTS

- **All India Rank 60** in the **14th National Interactive Mathematics Olympiad (NIMO)** among **14K+ students**.
- **National Top 1** Percentile in **National Standard Examination in Physics (NSEP) 2023** Conducted by **IAPT.**
- **Rank 3** in the **Optiver Trade-a-thon 2025** at **IIT Hyderabad**, among **2028 graduates**.
- **All India Rank 393** in the **IICPC QuantFest 2025 Prelims** among **10K+ participants**, **Rank 80** among **2028 graduates**.
- **All India Rank 318** in the **IICPC QuantFest 2025 Regionals** among **600 qualifiers** from the Prelims.
- **Global Rank 785** in the **IICPC CodeFest 2026 Prelims**, competing against **13,000+ top coders** worldwide.
- **All India Rank 3** in **IMC Prosperity 4** and **World Wide Rank 36** out of **20,000+** teams till last round.
- **All India Rank 197** in **AMS Derive PRIOR**, sponsored by **Jane Street** and **QRT**, among **1500+ top participants**.
- **Codeforces:Max Rating 1714 (Expert)**, **1,000+** problems solved (Handle: **0ne8**).
- **Grades: A+** in Introduction to Probability; **A** in Probability and Random Variables; **A** in Algorithms.
- **Campus Finalist** in **Integral Cup 2025**, sponsored by **Jane Street**.
- **Global Rank 2,902** in **Meta Hacker Cup 2025 Round 1** and **Global Rank 1,981** in **Round 2**, earning official Meta Hacker Cup merchandise.

RELEVANT COURSES

Mathematics: Introduction to Probability, Probability and Random Variables, Introduction to Statistics, Linear Algebra, Calculus, Discrete Mathematics, Number Theory.

Computer Science: Algorithms, Operating Systems, Database Management Systems, Computer Architecture, Theory of Computation, Introduction to Programming.

PROJECTS

Multi-Threaded Lock-Free Limit Order Book (C++) *Github*

- Built a **C++** multi-threaded **L3 Limit Order Book**, achieving **1.73M operations/sec** with an average latency of **577 ns** across **16 concurrent threads**.
- Developed a lock-free order book using a **Skip List**, **Michael-Scott Queue**, and **CAS-based synchronization** to support concurrent order insertion while preserving **FIFO** execution.
- Implemented **Epoch-Based Memory Reclamation (EBR)** to safely reclaim lock-free data structures while preventing **use-after-free** and **ABA** memory reclamation issues.
- Improved throughput by **87%** over a mutex-based baseline by eliminating **false sharing** and pinning worker threads to dedicated CPU cores.

Sparse Matrix Multiplication Accelerator (C++, Computer Architecture) *Github*

- Designed a custom **4-operand RISC-V instruction** for sparse matrix acceleration on a dedicated coprocessor.
- Implemented a **4-stage pipelined, 8-lane 64-bit SIMD** datapath for high-throughput sparse matrix multiplication.
- Engineered a **Row-Row multiplication algorithm** using **bit-packed offset metadata** to bypass zero values, completely eliminating redundant memory fetches and computation cycles.
- Reduced execution cycles**,** achieving **128× speedup at 75% sparsity and 320× speedup at 90% sparsity.**
- Validated correctness using a randomized C++ verification framework and projected **0.13 mm²** silicon area in **7 nm FinFET** technology.

xv6 Kernel Extensions for Scheduling, Virtual Memory & Storage Systems (C)

- Extended the **xv6 operating system** by implementing custom **system calls** for process introspection, parent-child tracking, and syscall accounting.
- Designed and implemented a **4-level Multi-Level Feedback Queue (MLFQ) scheduler** with priority boosting, preemption, and scheduling analytics.
- Developed a **clock-based page replacement algorithm** with frame tracking, page fault handling, and **disk-backed swap management**.
- Implemented **FCFS** and **SSTF** disk scheduling algorithms and added support for **RAID-0**, **RAID-1**, and **RAID-5** storage abstractions.

SKILLS

Programming Languages:C++ (Primary), Python, OCaml (Basic), Java (Basic), Bash, SQL

Tools & Technologies: Git, Linux, LaTeX

POSITIONS OF RESPONSIBILITY

Overall Head, Infero (Competitive Programming Club), IIT Hyderabad

Core Member, Infero (Competitive Programming Club), IIT Hyderabad

Quant Coordinator, Finance & Consulting Club (FCC), IIT Hyderabad

Internship Coordinator, Office of Career Services (OCS), IIT Hyderabad

EXTRACURRICULAR

4th Prize in **Gridlock**, a coding competition at IIT Hyderabad.

1st Prize in the Filmmaking Competition at **Milan**, IIT Hyderabad, as **Video Editor**.